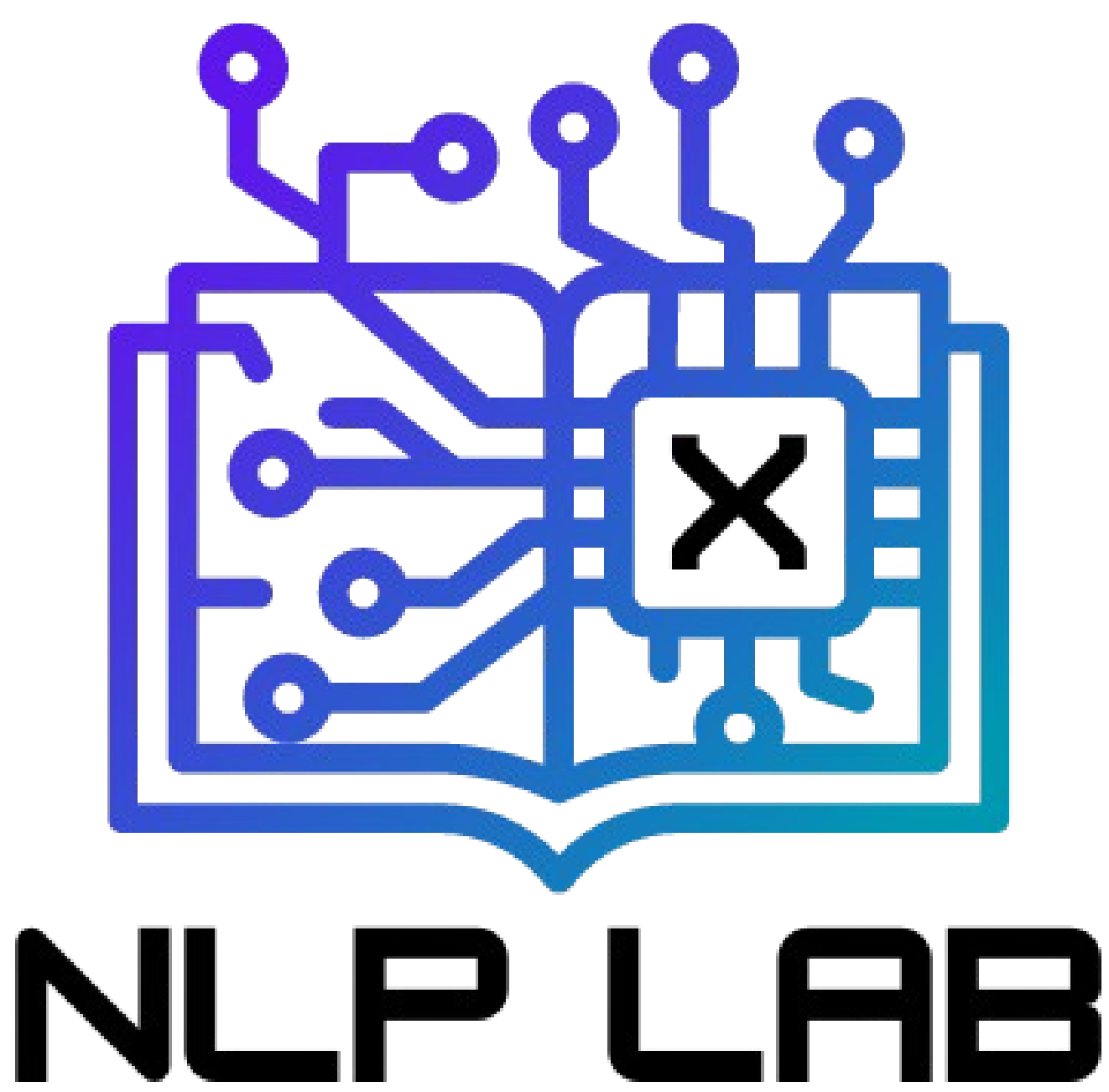


Explainability in Entity-Aware Financial Sentiment Classifiers

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TASK DESCRIPTION

Explain classification decisions made by an entity-aware sentiment analysis model using counterfactual explanations and feature attribution.

RESEARCH QUESTION

- How can we best use attribution-guidance to generate valid, plausible, and diverse counterfactual explanations in this context?
 - **Validity** A counterfactual is valid if it flips the class label of only the target entity while minimizing edits to the input.
 - **Plausibility** The CFE should present a scenario that is within the scope of possibility. A counterfactual is considered plausible if it exhibits a high degree of similarity to existing samples in the dataset. This is important to ensure the explanations provide recourse (actionable information) to the end-user.
 - **Diversity**: A set of counterfactuals are considered diverse if the token-distance between pairs of counterfactuals is high. This helps give insight into different parts of the input that could be modified to change the outcome.

DEFINITIONS

- Counterfactual explanation: A method of explaining an AI's decision using a counter example. Sounds like "If you had said it in this way I would classify differently"
- XAI: Explainable Artificial Intelligence
- LIME: Locally Interpretable Model Explanations
- Feature Attribution: The parts of the input that had the biggest effect on the prediction
- Entity-Aware: A model trained to predict the class of an entity (person/place/thing) $P(\text{class} | \text{text}, \text{entity})$
- Sentiment Analysis: Using algorithms to assign positive, negative, or neutral labels to text

CHALLENGES / MOTIVATION

- Financial text contains complex, domain-specific terminology; necessitating a deeper understanding of models that deal with it
- Samples often refer to multiple entities, causing confusion among most sentence-level models
- Language models used in high-stakes environments must be explainable in order to ensure trust and transparency

EXAMPLE

Counterfactual generation (first-shot, using Gemma3N: 4B) with and without feature attribution:

Original text: "Microsoft plunges, Meta rallies as investors demand AI payoffs" - Reuters

Entity: Microsoft **Class:** Negative

Feature Attribution

Plunges (0.89)	rallies (-0.11)	demand (0.06)
Microsoft (0.05)	payoffs (0.04)	AI (0.03)
investors (0.02)	as (0.02)	Meta (-0.01)

Feature Attribution: "Microsoft surges, Meta rallies as investors demand AI payoffs"

Baseline (No Attribution): "Microsoft surges, Meta dips as investors are satisfied with AI progress."

Criteria	Valid	Plausible	Minimally changed
With Feature Attribution	Yes	Yes	Yes
Baseline (No Attribution)	No	No	No

Interpretation: Using feature attribution, we can generate counterfactual explanations for these models that are more useful and effective

VISUALIZATION

Counterfactual Explanation (CFE)

